



NTU Academy for Professional
and Continuing Education

(SCTP) Advanced
Professional Certificate

Data Science and AI



3.2 Part 1: Introduction to Machine Learning

Module Overview

3.1 Probability and Statistics for Machine Learning

3.2 Introduction to Machine Learning

3.3 Supervised Learning

3.4 Supervised Learning - Advanced

3.5 Unsupervised Learning

3.6 Time Series Data & Forecasting

3.7 Neural Network & Deep Learning

3.8 Computer Vision (CV)

3.9 Natural Language Processing (NLP)

3.10 NLP - Advanced

Recap, Self-Study and Prework

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Lesson Objectives

- Understand the definition and different types of ML
- Compare the different data types
- Train linear regression, logistic regression and KNN models

Lesson Plan

- 3.2a: Introduction to Machine Learning
- 3.2b: Supervised Learning Algorithms
 - Linear Regression
 - Logistic Regression
 - K-Nearest Neighbours (KNN)

Agenda

- Introduction to Machine Learning
- Types of Machine Learning
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning
- Data Types
- Glossary

Recap on Terminology

Artificial Intelligence

Artificial Intelligence (AI) is an *umbrella term* for any technology where a computer program is attempting tasks that come naturally to the human brain.

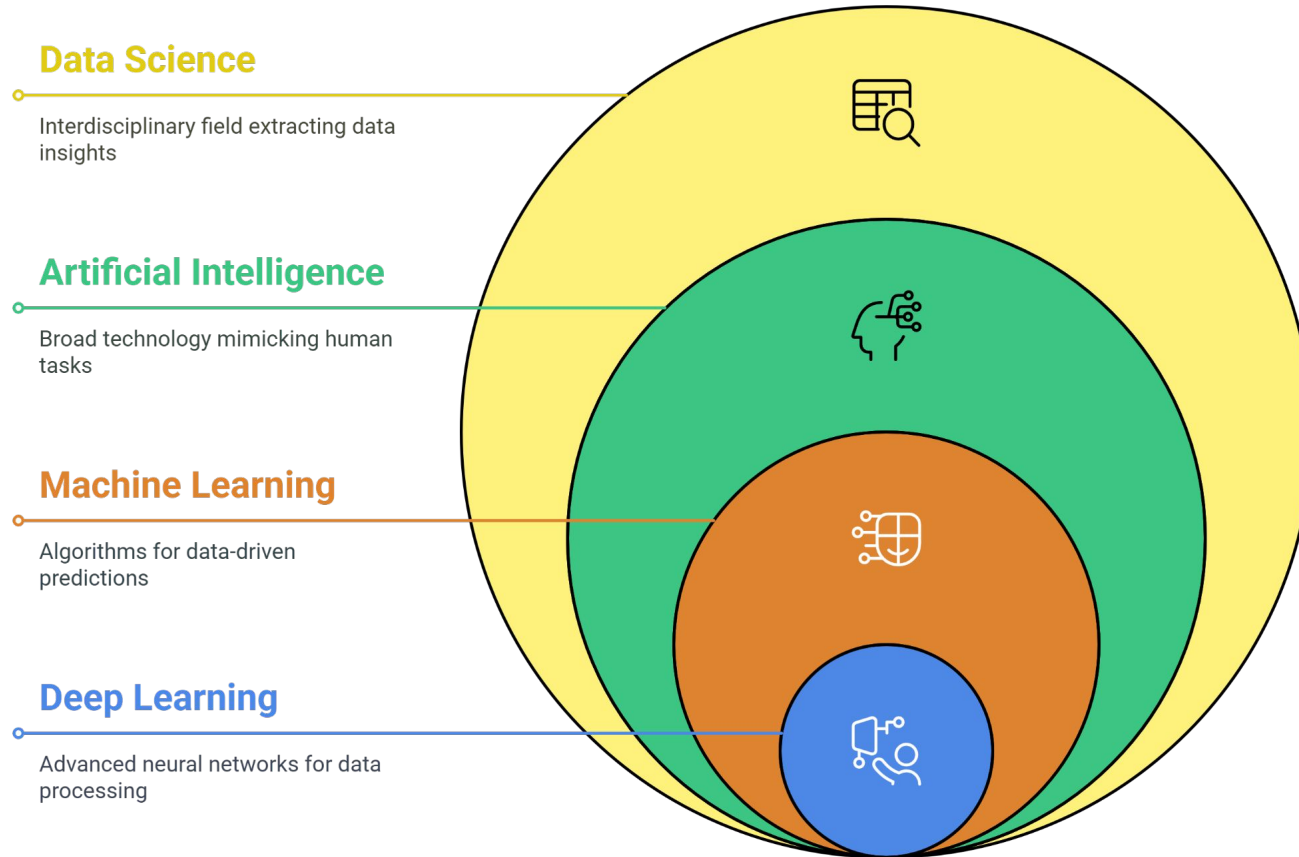
Machine Learning

Machine Learning (ML) is a *subset of AI* that involves training algorithms to make predictions or decisions based on data.

Data Science

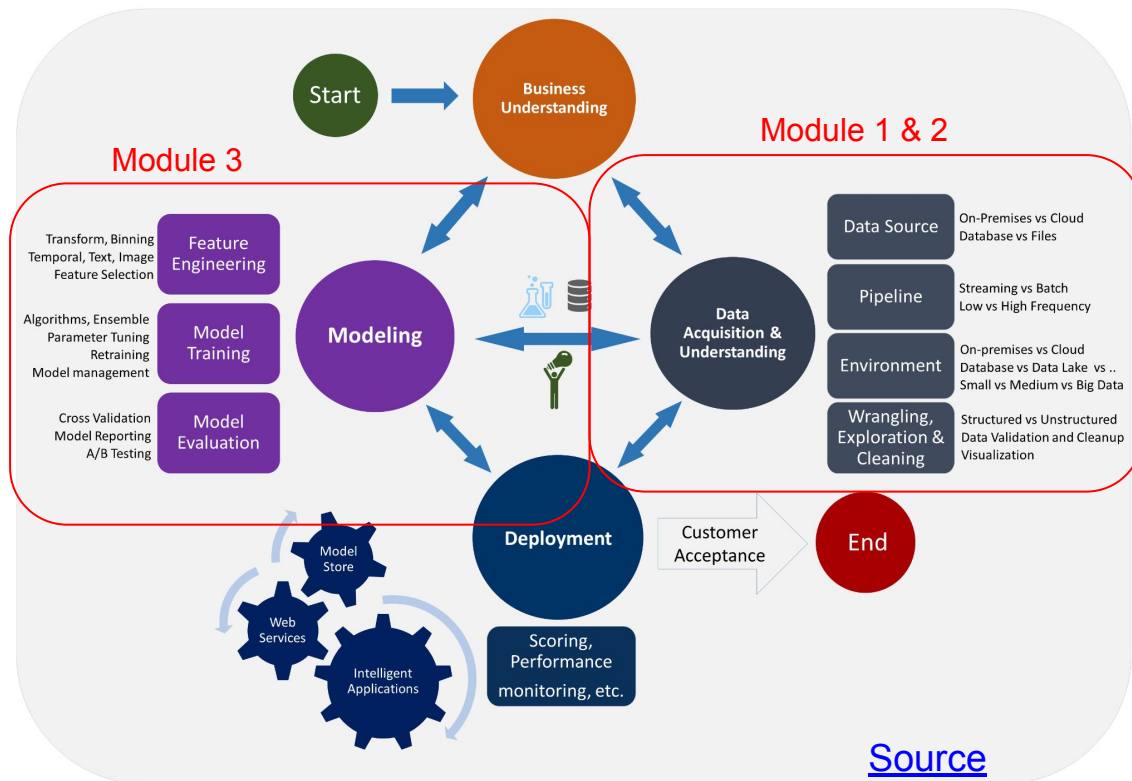
Data Science is an *interdisciplinary field* that uses statistics & ML methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data.

Hierarchy of Data Science and AI



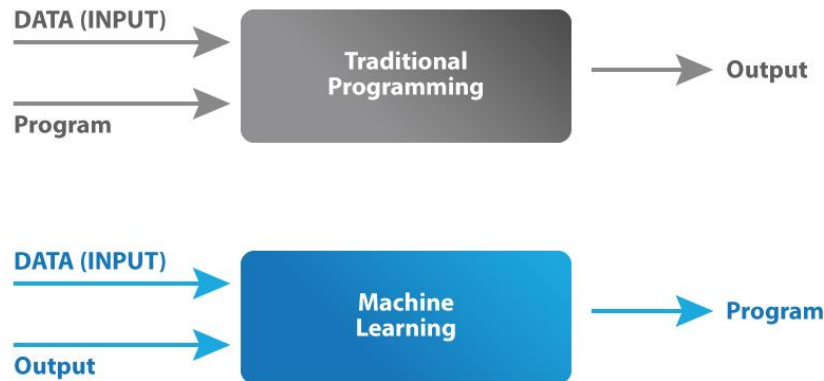
Recap on Data Science Lifecycle

1. Business understanding
2. Data acquisition and understanding
3. Modeling
4. Deployment
5. Customer acceptance



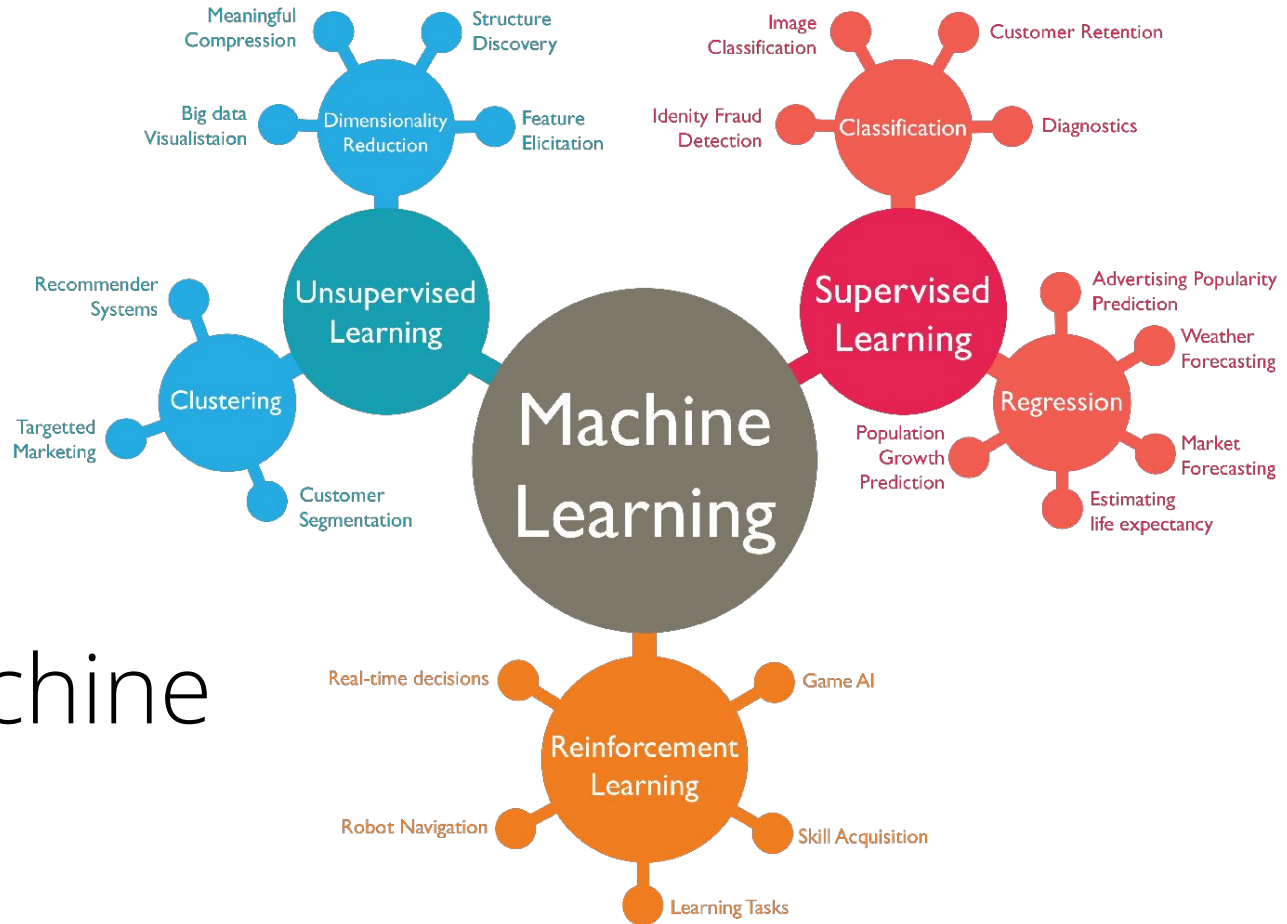
Introduction to Machine Learning

Machine Learning (ML) is a subset of artificial intelligence that provides systems the ability to *automatically learn and improve from experience* without being explicitly programmed.



Class Activity

What types of
Machine Learning
do you know?



Types of Machine Learning

Supervised Learning

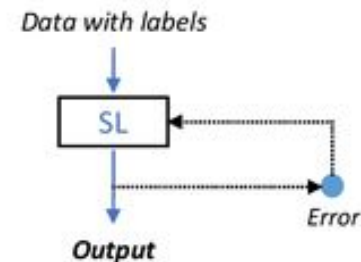
Supervised Learning involves training a model on a set of *input-output pairs*. It learns a function that *maps an input to an output* based on example input-output pairs.

Features & Labels:

- **Features** are the input; **Labels** are the output.
- Example: For an email spam detector, the email content is the feature, and the label is whether it's 'spam' or 'not spam'.

Training Process:

- A large dataset with labeled examples is provided.
- The algorithm iteratively makes predictions (*predicted output*) on the training data and they are compared with the labels (*actual output*).



Supervised Learning

Supervised Learning involves training a model on a set of *input-output pairs*. It learns a function that *maps* an input to an output based on example input-output pairs.

Features & Labels



Input (Feature)
e.g., email content



Output (Label)

Training Process



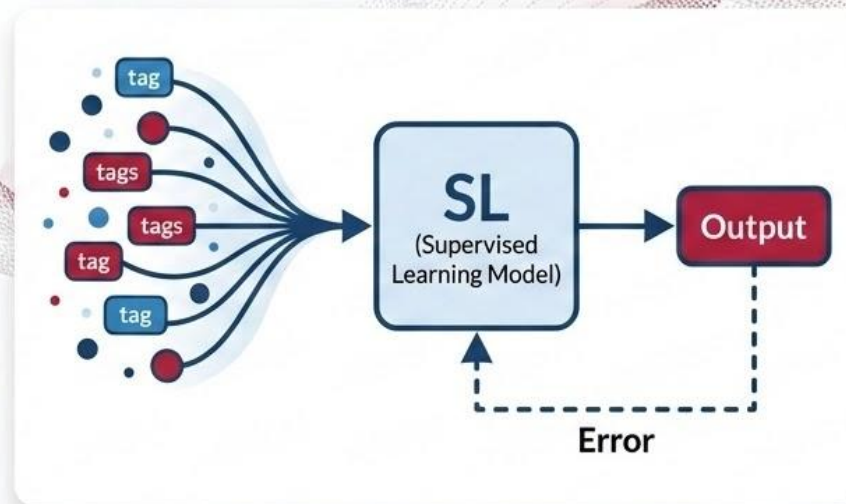
Labeled
Dataset



Algorithm Makes
Predictions



Compare with
Actual Labels (Error)



Unsupervised Learning

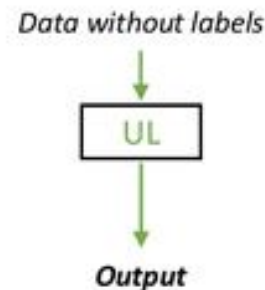
Unsupervised Learning involves training a model on a set of data *without explicit instructions* on what to look for. The goal is to *identify patterns* or *structures* in the data.

Clustering & Dimensionality Reduction:

- Clustering groups similar data points together.
- Dimensionality Reduction reduces the number of random variables under consideration by obtaining a set of principal variables.

Training Process:

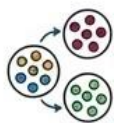
- The algorithm explores the data for patterns.
- There's no 'correct answer' (labels) like in supervised learning.



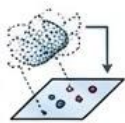
Unsupervised Learning

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Clustering & Dimensionality Reduction



Clustering groups similar data points together.



Dimensionality Reduction reduces the number of random variables under consideration by obtaining a set of principal variables.

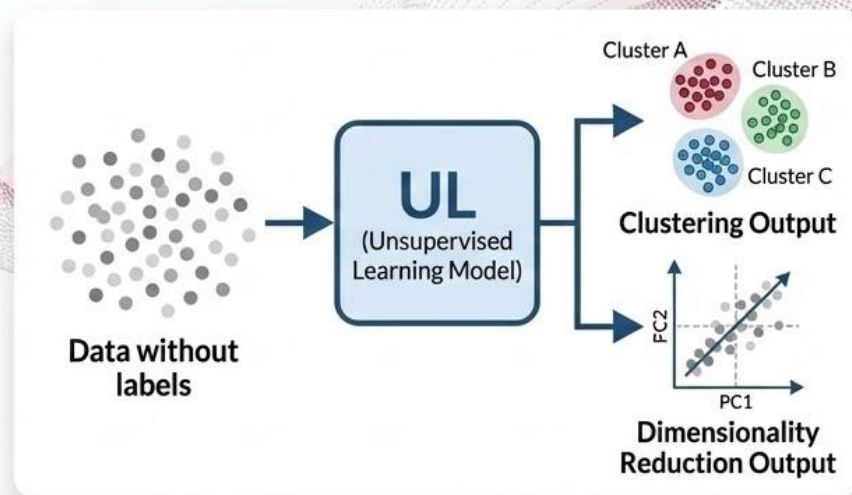
Training Process



The algorithm explores the data for patterns.



There's no 'correct answer' (labels) like in supervised learning.



Reinforcement Learning

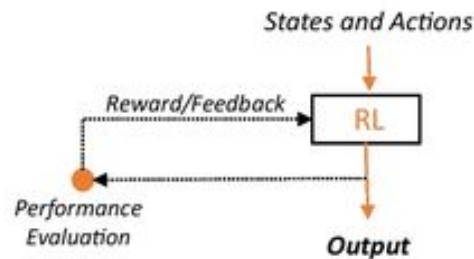
Reinforcement Learning is about taking suitable actions to *maximize reward* in a particular situation. It's trained using *trial and error*.

Agent, State, Reward:

- **Agent:** The decision maker.
- **State:** The current situation returned by the environment.
- **Reward:** Feedback from the environment.

Training Process:

- The agent takes actions; the environment returns the new state and a reward.
- The goal is to find the best strategy, called **policy**, that will result in the maximum cumulative reward for the agent.



Reinforcement Learning

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State

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Reward

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Training Process

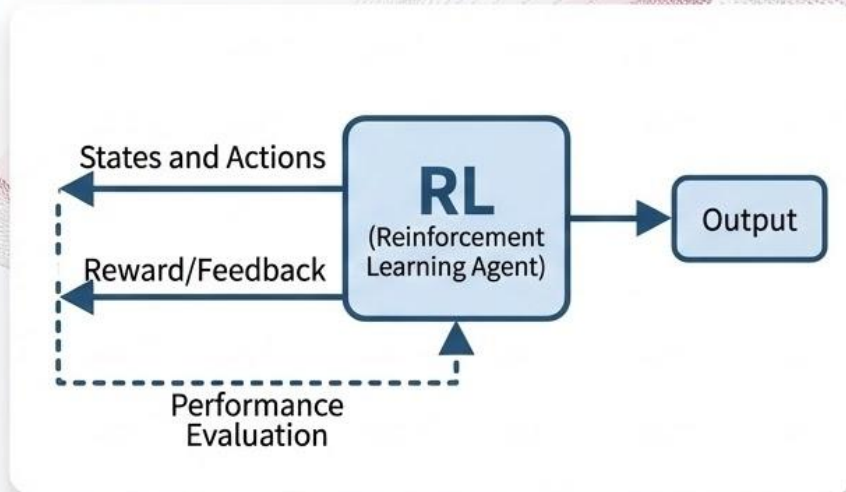


Trial and Error

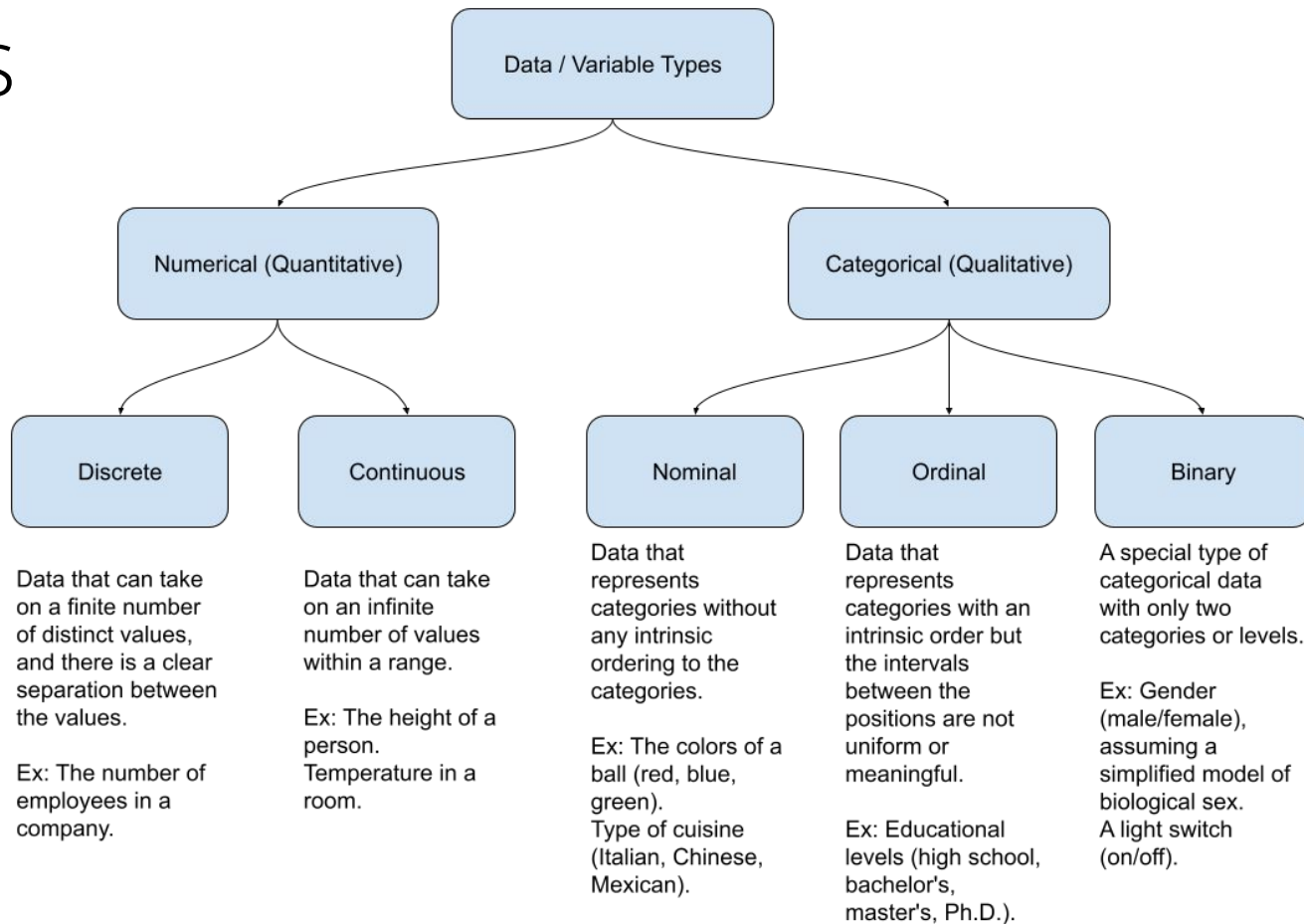


The agent takes actions; the environment returns the new state and a reward.

The goal is to find the best strategy, called **policy**, that will result in the cumulative reward for the agent.



Data Types



Data Types

Aspect	Time Series Data	Cross-Sectional Data
Definition	Data collected or recorded at a regular interval over time.	Data collected at a single point in time or over a very short period.
Dimensionality	One-dimensional (time)	Multidimensional (multiple variables or entities)
Examples	Stock prices over months/years	Survey data on income levels across different households
	Daily temperature readings	Cross-sectional study of a disease in a population
Analysis Focus	Trends over time	Relationships or differences across groups/entities
	Seasonal patterns	Snapshot of a particular time
Data Structure	Typically organized in chronological order	Organized by entity, with various attributes for each

Glossary

- Variable:** An attribute that represents a dimension of the data. Variables can be used as input features for models, and they can also be the output or target that the model is trying to predict. Variables can be categorical or numerical.
- Feature:** An individual measurable property or characteristic of a phenomenon being observed, often denoted as 'X' in models. Also known as *input*, *independent* or *predictor* variable.
- Label:** The variable we are predicting, often denoted as 'Y' in models. Also known as *output*, *dependent* or *target* variable.
- Algorithm:** A set of rules or procedures that a machine learning system follows to perform a task.
- Model:** The representation (internal model) learned by the machine learning algorithm.

Supervised Learning (Types)

Aspect	Regression	Classification
Definition	Regression predicts a numerical output.	Classification predicts a categorical output.
Output Type	Numerical values (e.g., prices, temperatures).	Categorical labels (e.g., spam/not spam, dog/cat).
Nature of Predictions	Predicts a quantity that can take any value within a range.	Predicts a class label or category for the observation.
Algorithms	Linear Regression, Decision Trees, Random Forest, Support Vector Machines, Neural Networks, etc.	Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, Neural Networks, etc.

Supervised Learning (Examples)



Regression	Classification
Predicting the price of a house based on its size, location, etc.	Determining if a tumor is benign or malignant based on medical imaging.
Forecasting tomorrow's temperature	Email spam detection